



OpenSCUBA

An open-source 3D SCUBA diving experience

Interactive Graphics

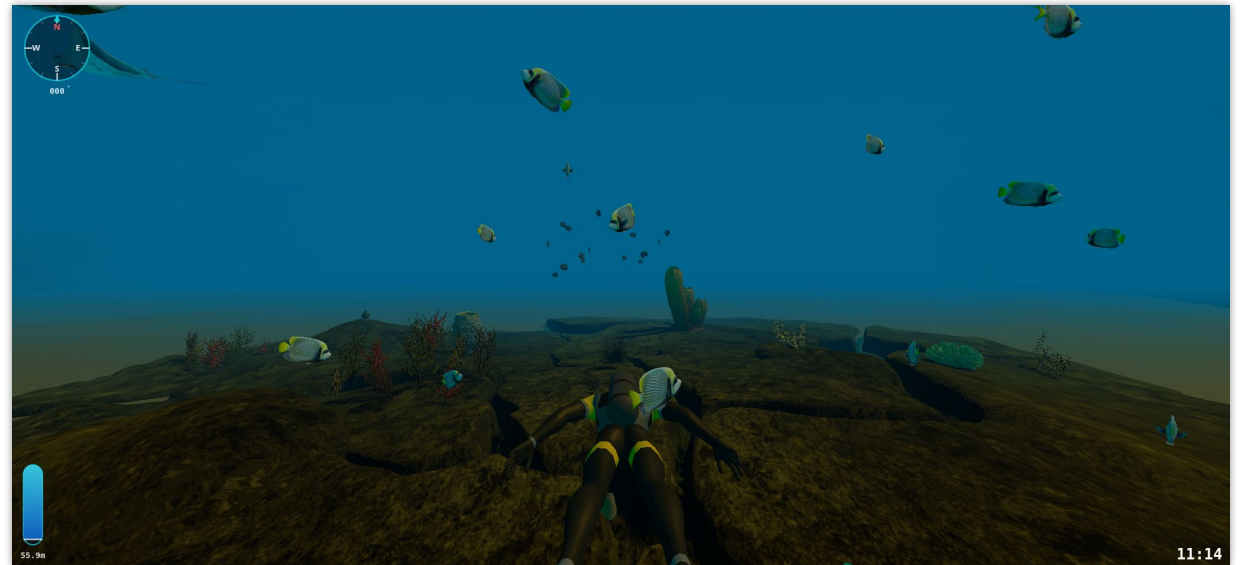
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Project Overview

OpenSCUBA is an interactive underwater exploration experience, inspired by ARIKA's *Endless Ocean* series and its relaxing underwater atmosphere. It runs entirely in the browser and is fully open source on GitHub.

Features:

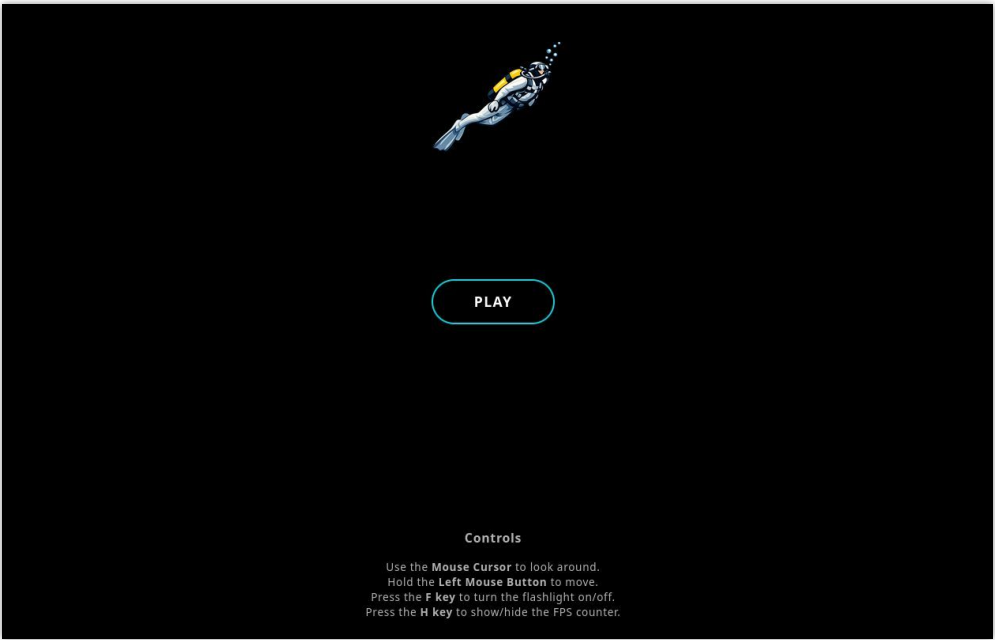
- **Immersive underwater setting** with varied scenery;
- User-controllable **scuba diver character**;
- **Marine creatures** of different sizes;
- Heads-Up Display (**HUD**);
- **Sound effects** and **music**.



How to Run

The live demo runs directly in a web browser via GitHub Pages: no installation or configuration required.

A Loading UI appears first; once all assets have finished loading, a “Play” button starts the experience.



Controls

Input	Action
Mouse cursor	Look around the environment
Left mouse button (hold)	Move the diver
F key	Toggle flashlight on / off
H key	Show / hide FPS counter

Environment and Libraries

Built on Three.js, with:

- **simplex-noise** → Procedural noise for the seabed;
- **three-mesh-bvh** → Accelerated spatial queries, collisions and raycasts;
- **tween.js** → Smooth animations and transitions.

Build tooling

- **vite** → Builds the environment;
- **vite-plugin-static-copy** → Copies static assets to the build output.

Deployment

- A GitHub Actions workflow automatically builds and deploys the app on every push.

Note: all 3D models, textures and sounds are either third-party or AI-generated. See the repository README file for details.

Project Structure

Project folders are organized to prioritize clarity and modularity:

assets/

3D models, organized into subfolders

components/

Scene element classes (water, sun, creatures, ...)

core/

Main scene files and controls configuration

docs/

Project documentation

shaders/

GLSL shader classes

sounds/

Sound effects and music files

textures/

Texture files

ui/

UI behaviour classes

utils/

Reusable utility classes

Core Architecture

World Setup

- A World class (`core/world.js`) instantiates the scene, renderer, camera and lighting;
- **Global soft shadow mapping** via Percentage-Closer Filtering (PCF);
- A render function updates dynamic components every frame;
- The `main.js` script instantiates World and is executed by `index.html`;
- **Application Workflow:**
 - a. Loading UI;
 - b. Play button;
 - c. Main scene enter and controls enabled (mouse look and movement); Controls are paused on loss of window focus or out-of-bounds cursor.

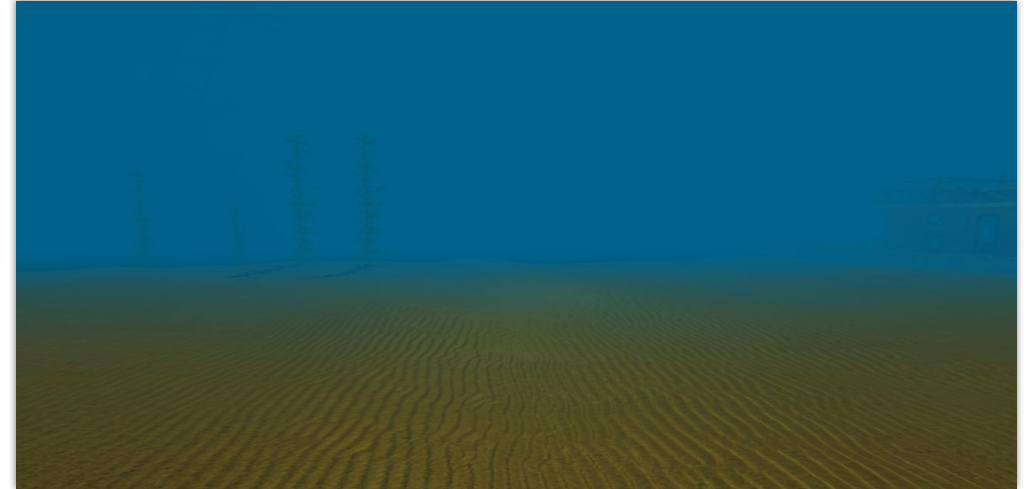
Resource Management

- GarbageCollector utility disposes meshes, materials and textures to limit memory leaks;
- WebGL renderer requests “powerPreference: 'high-performance'” to favor the dedicated GPU on hybrid laptops.

Underwater Environment

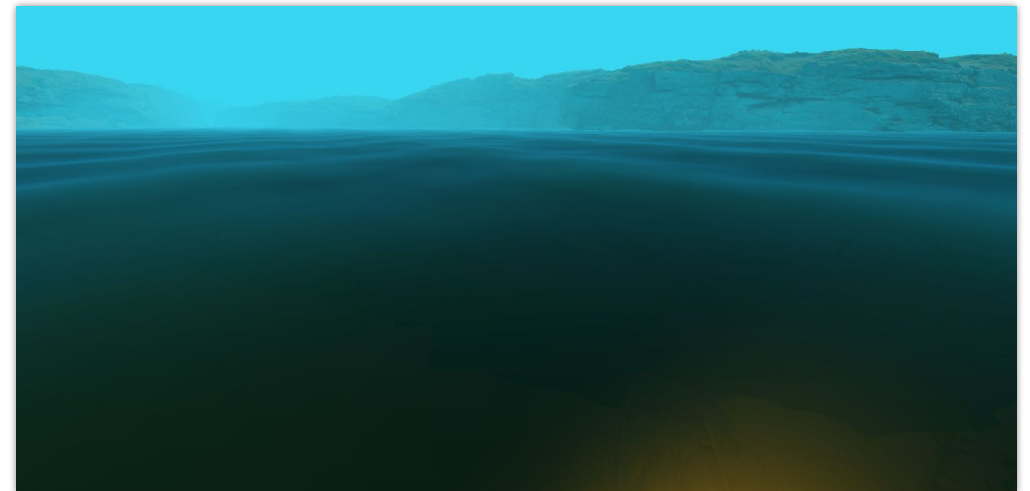
Seabed

- 800×800 PlaneGeometry deformed with simplex-noise;
- Normals recomputed for correct lighting on the deformed surface;
- Sand-colored material with soil + wavy-sand textures (RepeatWrapping);
- Configured to receive shadows.



Water

- 800×800 PlaneGeometry with a custom GLSL shader;
- Cyan ambient light + exponential fog create the underwater feel.



Water Shader

Vertex Shader → Gerstner waves

Wave	Steepness	Wavelength	Role
1	0.05	20.0	Broad primary swell
2	0.035	14.0	Secondary swell
3	0.02	9.0	Fine surface detail
4	0.01	6.0	Fine surface ripple

Four waves are summed for realistic ocean motion; amplitude (A) is derived from steepness (S) and wavelength (L). Normals are recomputed numerically each frame from nearby sampled points.

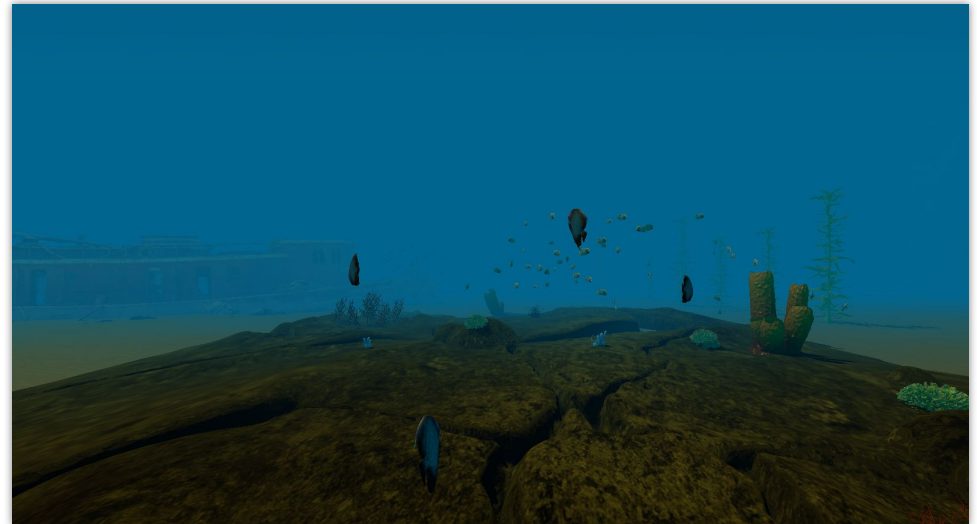
$$A = \frac{SL}{2\pi}$$

Fragment Shader and Depth Effect

- Approximated **Fresnel reflection**, capped at 70% to avoid a mirror-like surface;
- **Pixel coloring** based on surface elevation;
- **Blinn-Phong lighting** blended with the sky color;
- Fog color and density interpolate with camera height, simulating depth.

Sun and Day-Night Cycle

- The sun is a SphereGeometry with an attached shadow-casting DirectionalLight;
- A subtle glow, generated with a CanvasTexture, is applied to the sphere;
- An update routine keeps the light and glow following the sun every frame;
- Sun position, sun color and the renderer's clear color all evolve on a day-night cycle;
- A full day lasts 1200 seconds (= 20 minutes).



Scenery Elements

Rock walls

Define the world boundaries realistically

Rock mountain

Features a custom color texture

Shipwreck

A major underwater landmark

Artificial reef

An additional man-made structure

Coral reef

Base rock + coral models, placed via SceneryWrapper

Kelp plants

Batched placement; collisions disabled for realism trade-off

Machinery

Futuristic device with a night-time light beam (SpotLight + PointLight, animated via tween.js)



Scuba Diver

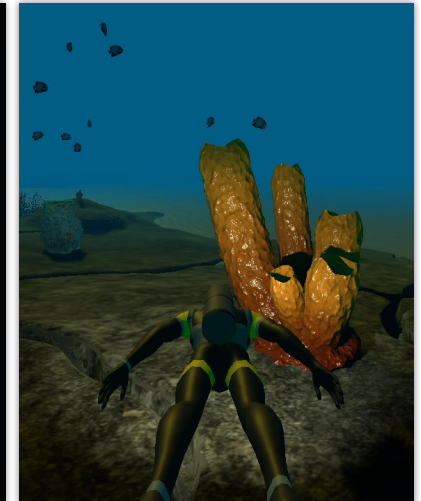
Model and Animation

- Third-party Mixamo-style rigged model; all animation was written from scratch;
- The bind (rest) pose is reconstructed from the skeleton's inverse bind matrices;
- *Idle* and *swimming* states are blended via a smoothed factor;
- Limb motion is driven by offset sine/cosine cycles, mirrored left/right.



Camera, Controls and Flashlight

- Mouse look + LMB hold-to-move → The diver follows the camera in third person;
- The diver's head independently rotates toward the cursor;
- **Flashlight** (F key): SpotLight, intensity 100, range 30, 60° cone, casts dynamic shadows.



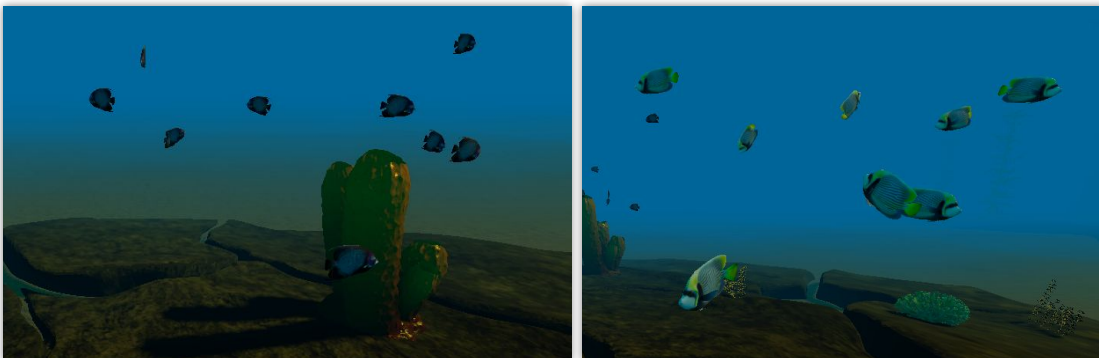
Collision Detection

- **Creature Collisions** (`resolveCreaturePushback`):
 - Padded axis-aligned bounding box per creature; the diver is pushed out along the axis of minimal overlap.
- **Scenery Collisions** (`resolveSceneryPushback`):
 - Closest-surface-point check using three-mesh-bvh bounds trees; the diver is pushed out along the surface normal.
- **Environment bounds**:
 - A downward Raycaster keeps the diver above the (noisy) seabed;
 - A Y-position clamp keeps the diver below the water surface.

Marine Creatures

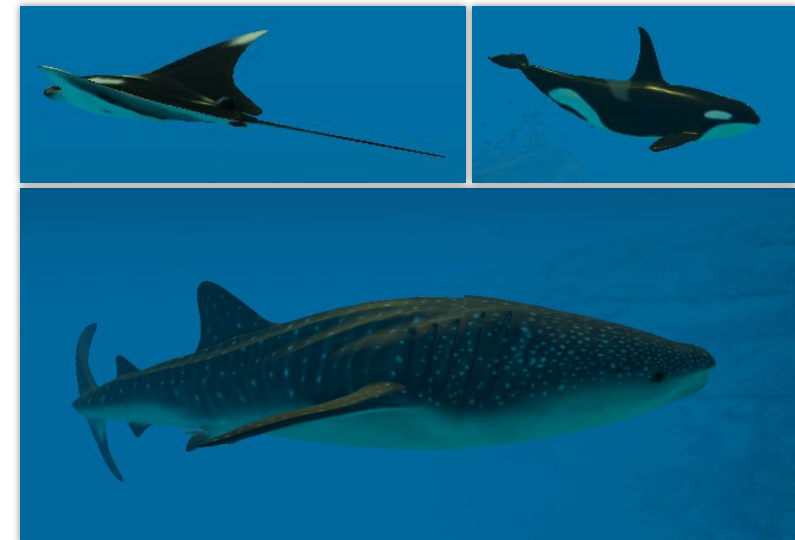
Small Creatures (smaller than the diver)

- **Random wandering algorithm** (`_updateWander`):
 - a. Idle state;
 - b. New destination pickup (`_pickNewWanderTarget`);
 - c. Travel with deceleration.
 - d. Smooth turn (`slerp`);
- **Available Creatures:**
 - a. **Coral Fish**;
 - b. **Emperor Angelfish**.



Large Creatures (larger than the diver)

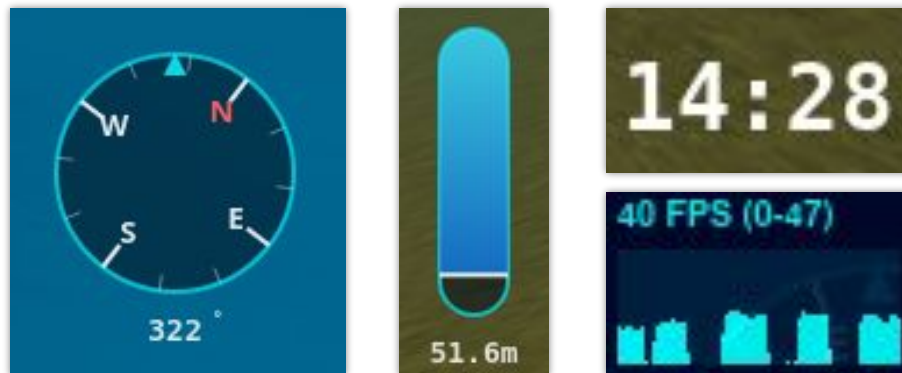
- Hard-coded trajectories, unique to each species;
- **Available Creatures:**
 - **Whale Shark** → Circular path, swinging tail fin;
 - **Manta Ray** → Ellipsoidal path, flapping pectoral fins;
 - **Killer Whale** → Linear path, oscillating tail fin.



HUD and Audio

HUD elements

- **Compass** (top-left) → Camera yaw, graphic + degrees;
- **Depth Gauge** (bottom-left) → Diver depth, graphic + meters;
- **Time Display** (bottom-right) → World time in <hh:mm> format;
- **FPS Counter** (top-right) → Toggled with H, hidden by default.



Sound and Music (AudioManager)

- **Play-button** click, followed by a **splash** on scene transition;
- **Underwater ambience** loop;
- **Diver breathing**, low volume, positional;
- **Flashlight** click sound;
- **World music**, played once after entering the scene.

Hands On!

Let's see *OpenSCUBA* in action!

Conclusion

- **Core Experience:** Browser-based 3D underwater simulation combining an explorable environment, HUD, and audio.
- **Key Mechanics:** Features a procedurally animated diver alongside wandering and path-following marine creatures.
- **Technical Highlights:**
 - Custom multi-wave water shader.
 - Blended idle-to-swimming character animations.
- **Potential Future Extensions:**
 - Expanded fauna and scenery.
 - New biomes (e.g., deep abyss, freshwater rivers).
 - Enhanced diver-environment interactions.

An underwater scene with a blue background. Two divers are swimming in the upper half of the frame. The diver on the left is wearing a black wetsuit with a yellow tank, and the diver on the right is wearing a white wetsuit with a yellow tank. Both are exhaling bubbles. In the foreground, a diver is seen from behind, swimming towards the bottom. The bottom is covered in brown sand and rocks, with some green and red coral or seaweed. Several small fish are swimming around, and a larger green fish is on the right side.

Thanks for your attention!